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Digitalization and Sustainable Industrial Low-Carbon Transformation: Synergistic Effects, Policy Tools, Technical Pathways, and Financial Innovation

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Abstract

In the context of the growing urgency of sustainable industrial transformation under global climate goals, this study examines how digitalization enables and amplifies industrial low-carbon transition through the synergistic interaction of policy tools, technological pathways, and financial innovation. Addressing the challenge of reconciling emissions reduction with industrial efficiency, the study employs a mixed-method approach that combines panel econometric analysis of manufacturing enterprises in China's Yangtze River Delta with representative case studies. The empirical results demonstrate significant synergistic effects among policy, technology, and finance under digital enablement. Coordinated policy instruments, including emissions trading and green credit, reduce decarbonization costs by 18–23%, while digitally enabled mechanisms such as Zhejiang's "Carbon Efficiency Code" lower carbon intensity by over 15% for nearly half of participating firms. Technological pathways exhibit sectoral heterogeneity: digital twin optimization reduces emissions by 12% in the steel industry, whereas IoT-based monitoring cuts energy consumption by 9.7% in textiles. Financial innovations further reinforce these outcomes by increasing green R&D intensity and enhancing firms' climate risk resilience. From a sustainability perspective, the study shows that digitalization strengthens real-time carbon measurement, monitoring, and verification (MRV), thereby improving sustainability performance assessment and governance effectiveness. By integrating digital tools with policy and financial incentives, the findings provide actionable guidance for supporting sustainable industrial operations and designing more precise, scalable, and data-driven sustainability-oriented policy instruments.

Keywords: sustainability performance; sustainable operations; environmental governance; policy tools; technological pathways

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1. Introduction

Global climate change has emerged as one of the most severe challenges confronting human society in the 21st century. Reports from the Intergovernmental Panel on Climate Change (IPCC) consistently emphasize that achieving global net-zero emissions by mid-century is imperative to avert irreversible catastrophic consequences of climate change. Against this backdrop, nations worldwide have elevated climate action to the level of national strategy, actively setting carbon peak and carbon neutrality targets. As the world's largest developing country and carbon emitter, China solemnly pledged in 2020 to strive for carbon peaking by 2030 and carbon neutrality by 2060 (referred to as the "dual carbon" goals) [1]. This ambitious target not only demonstrates China's responsibility as a major power but also heralds profound transformations for its socio-economic development, particularly within the industrial sector.

The industrial sector, a pillar of the national economy, is also the primary source of energy consumption and greenhouse gas emissions. According to data from the National Bureau of Statistics, China's industrial value-added has long maintained a high proportion of GDP, with industrial energy consumption and carbon emissions accounting for the vast majority of the national totals. Consequently, the green and low-carbon transformation of industry is pivotal to achieving the "dual carbon" goals. However, industrial transformation faces numerous challenges, including inertia in high-carbon industrial structures, substantial capital requirements for technological upgrades, path dependency on traditional production models, and insufficient green technology innovation. Effectively promoting deep decarbonization of the industrial sector while ensuring economic development has become a critical theoretical and practical issue demanding urgent resolution.

Amidst the global consensus on climate action, the rise of the digital wave presents unprecedented opportunities for industrial low-carbon transformation. Digital technologies, represented by big data, artificial intelligence, the Internet of Things (IoT), cloud computing, and digital twins, are profoundly altering the organization, operation, and value creation of industrial production. These technologies enable precise monitoring and optimization of energy consumption, intelligent control of production processes, green management of supply chains, and real-time accounting and forecasting of carbon emissions, thereby significantly enhancing the energy and resource utilization efficiency of industrial production and reducing carbon intensity. For example, through industrial internet platforms, enterprises can collect and analyze equipment operational data in real-time, identify and optimize abnormal energy consumption points, achieving refined management for energy saving and emission reduction [2]. Digital twin technology can create virtual replicas of the physical world, simulating and optimizing production processes to minimize waste and energy consumption.

However, digital technologies do not exist in isolation; their role in driving industrial low-carbon transformation requires effective policy guidance and adequate financial support. Governments provide external incentives and constraints for industrial enterprises' low-carbon transition by formulating and implementing diverse policy tools such as carbon emissions trading, green credit, fiscal subsidies, and technical standards. For instance, carbon emissions trading mechanisms price carbon emissions through market-based instruments, incentivizing companies to reduce emissions when marginal abatement costs are lower than the carbon price. Green credit and subsidy policies provide financial support for technological upgrades and green projects. Concurrently, the financial sector is actively innovating, developing green financial products like green bonds, carbon efficiency loans, and green insurance, offering diversified financing channels and risk management tools for the low-carbon transition. These policy and financial innovations, together with digital technologies, constitute the three pillars driving industrial low-carbon transformation.

Despite the crucial roles played by policy tools, technical pathways, and financial innovation in industrial low-carbon transformation, the interactions and synergistic effects among them remain underexplored. Existing research predominantly focuses on the impact of single factors on low-carbon transition or explores relationships between two factors (e.g., the carbon reduction effects of "innovation-finance" policy combinations [3]), lacking systematic theoretical construction and empirical testing of how the three synergize and the enabling role of digitalization within this synergy. For example: How does digitalization amplify the incentive effects of policies? How does financial innovation accelerate the application of digital green technologies? These questions demand in-depth investigation. Understanding this multidimensional synergistic mechanism holds significant theoretical and practical importance for formulating more effective and targeted policies, guiding enterprises towards efficient transformation, and enabling financial institutions to better serve green development.

This study aims to fill the gap in multidimensional synergy analysis within existing research by constructing a comprehensive analytical framework to deeply explore how policy tools, technical pathways, and financial innovation interact under the backdrop of digitalization to collectively promote the low-carbon transformation of the industrial sector. Specifically, the study will:

1. Systematically identify and analyze the key policy tools driving industrial low-carbon transformation, digital technology pathways, and green financial innovation models, elucidating their respective mechanisms of action.
2. Construct a theoretical model of the multidimensional synergistic effects among policy tools, technical pathways, and financial innovation, incorporating the enabling role of digitalization.
3. Conduct empirical analysis using multi-source heterogeneous data on manufacturing enterprises in the Yangtze River Delta region, quantitatively assessing the impact of different synergistic models on industrial carbon emission intensity, energy efficiency, and green innovation capability.
4. Delve into the intrinsic mechanisms of synergy and validate the theoretical model and empirical results through in-depth case studies.
5. Propose actionable policy recommendations to provide decision-making references for governments, enterprises, and financial institutions in promoting industrial low-carbon transformation, thereby contributing to the achievement of the "dual carbon" goals.

The innovations of this study lie in: First, it constructs a four-dimensional synergistic analytical framework of "policy–technology–finance–digital enablement," transcending the limitations of traditional single- or dual-factor analyses. Second, it emphasizes digitalization as a bridge and amplifier between policy/financial tools and low-carbon transformation, revealing the intrinsic mechanism of digitally enabled synergistic effects. Third, through empirical analysis of the Yangtze River Delta manufacturing sector, it provides quantitative evidence and validation of multidimensional synergistic effects, enhancing the practical guidance of the research. The findings of this study will offer novel theoretical perspectives and practical paradigms for the green and low-carbon transformation of the industrial sector in China and globally.

2. Materials and Methods

2.1. Digital Enablement for Low-Carbon Transformation: Theoretical Foundations and Practical Progress

Digital technology, as the core driver of the new wave of scientific and technological revolution and industrial transformation, is profoundly reshaping the global landscape of economic and social development. Against the backdrop of addressing climate change, the deep integration of digitalization and low-carbon transformation has given rise to the emerging research field of "digital greening". The theoretical foundation for digital enablement in low-carbon transformation stems primarily from its unique advantages in information transmission, optimized resource allocation, and efficiency enhancement [4]. Digital technologies can significantly improve energy utilization efficiency, optimize production processes, promote industrial structure upgrading, and ultimately reduce carbon

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emission intensity [5,6]. Specifically, digital enablement for low-carbon transformation manifests mainly in the following aspects:

1. Energy Efficiency Enhancement: Digital technologies enable refined management of energy consumption through real-time monitoring, data analysis, and intelligent control [7,8]. For example, IoT sensors can collect equipment energy consumption data in real time; big data analytics can identify abnormal energy consumption and waste points, suggesting optimizations. Artificial intelligence can predict energy demand, optimize energy dispatch, and reduce energy waste. The construction of smart grids enhances the operational efficiency of power systems and the integration capacity for renewable energy. Studies indicate that digital technological innovation effectively reduces urban carbon emission intensity by lowering energy consumption intensity [9].
2. Production Process Optimization: Technologies like digital twins and industrial internet can create virtual replicas of the physical world, simulating and optimizing production processes to reduce material consumption and waste generation. For instance, in high-energy-consumption industries like steel and chemicals, digital twin technology can simulate blast furnace operations and reaction processes, optimizing parameter settings to lower energy use and emissions. Industrial internet platforms enable interconnectivity among production equipment, optimizing production scheduling and coordination, enhancing production efficiency, and reducing energy waste caused by unplanned downtime.
3. Industrial Structure Upgrading: Digital transformation can propel traditional high-carbon industries towards low-carbon, high-value-added industries and foster new green industries. Digital technologies promote the development of new business models such as service-oriented manufacturing and the sharing economy, reducing material resource consumption. Simultaneously, digital technologies provide platforms and tools for green technology innovation, accelerating the growth of green industries like clean energy and energy conservation. For example, digital technologies play a crucial role in driving structural changes within the energy industry and promoting its low-carbon green development [10].
4. Carbon Emission Management and Accounting: Technologies like blockchain and big data provide technical support for precise Monitoring, Reporting, and Verification (MRV) of carbon emissions. Blockchain technology can establish a transparent and traceable carbon emission data management system, enhance data credibility and ensure the effective operation of carbon trading markets and carbon inclusive mechanisms. Big data analytics can uncover patterns in carbon emissions from vast datasets, providing a scientific basis for policymaking and corporate emission reduction [11].

Despite the immense potential of digital enablement for low-carbon transformation, challenges remain, such as the high costs of digital infrastructure development, data security and privacy concerns, the digital divide, and the need for coordination between digital and green transformations. Therefore, further research is needed on how to better leverage the enabling role of digitalization and address its potential issues.

2.2. The Role of Policy Tools in Low-Carbon Transformation: Classification, Mechanisms, and Combination Effects

Governments play a central role in promoting low-carbon transformation by designing and implementing diverse policy tools to guide and incentivize participation in emission reduction across society. Policy tools are typically classified into command-and-control, market-based, and incentive-based (or fiscal) types [12].

1. Command-and-Control Policies: These policies directly regulate corporate behavior through mandatory standards, regulations, and limits, such as carbon emission caps, energy efficiency standards, and the phasing out of obsolete production capacity. Their strengths lie in strong enforceability and rapid results, but they may lack flexibility, potentially leading to "one-size-fits-all" approaches and increasing corporate compliance costs [13].
2. Market Based Policies: These policies guide corporate emission reduction through market mechanisms, with the Emissions Trading System (ETS) being the most representative. ETS prices carbon emissions, incentivizing companies to reduce emissions when marginal abatement costs are lower than the carbon price, thereby achieving emission reduction targets at lower costs. Research shows that carbon emission trading policies can promote corporate digital transformation and facilitate their shift towards low-carbon pathways in pilot regions by fostering digital technology innovation and reducing financing costs [14].
3. Incentive-Based (Fiscal) Policies: These policies encourage corporate emission reduction actions by providing economic incentives, such as fiscal subsidies, tax incentives, green credit, and green bonds. Fiscal subsidies directly lower the costs of green technology upgrades for enterprises; tax incentives reduce the tax burden on green investments; green credit and green bonds provide low-cost financing channels. These policies effectively alleviate the economic burden of low-carbon transformation for enterprises and stimulate their motivation for emission reduction.

The effectiveness of a single policy tool is often limited and may have drawbacks [15]. For instance, command-and-control policies may stifle innovation, market-based policies may face price volatility risks, and incentive-based policies may involve fiscal burdens. Consequently, combining different types of policy tools to leverage their synergistic effects has become a significant direction in policy research. For example, the "innovation-finance" policy tool combination has been identified as having a significant impact on urban carbon emissions, providing policy recommendations for achieving the "dual carbon" goals [3]. This combinatorial effect can compensate for the shortcomings of single policies, forming a more robust and resilient policy system to more effectively drive low-carbon transformation.

2.3. Technical Pathways for Industrial Low-Carbon Transformation: Key Technologies and Application Scenarios

Technological innovation is the fundamental driver of industrial low-carbon transformation [16]. With the advancement of digital technologies, a series of key technologies have emerged, providing new pathways for industrial decarbonization. These technologies can be broadly categorized into process optimization technologies, end-of-pipe treatment technologies, and crosscutting enabling technologies [17].

1. Process Optimization Technologies: These technologies aim to reduce energy consumption and carbon emissions by improving production processes and equipment efficiency. Examples include: (1) Energy Management Systems (EMSs)—digital EMSs can monitor, analyze, and optimize energy consumption in real time, identifying waste and improving efficiency. (2) Advanced Control Systems—these are intelligent control systems built on artificial intelligence and big data optimize production parameters, reducing material and energy waste while improving product quality. (3) Digital Twin Technology—by creating virtual models of physical assets, digital twins enable real-time monitoring, simulation, and optimization of production processes, significantly reducing energy and resource consumption. For example, in

the steel industry, digital twins optimize blast furnace operation, lowering energy intensity and carbon emissions.

2. End-of-Pipe Treatment Technologies: These technologies focus on capturing and utilizing generated carbon emissions. Examples include: (1) Carbon Capture, Utilization, and Storage (CCUS)—CCUS technology captures CO2 from industrial emissions for permanent storage or use in other industrial processes. (2) Waste Heat Recovery—digital monitoring and control systems optimize waste heat recovery processes, converting waste heat into usable energy, thereby reducing overall energy consumption.
3. Cross-Cutting Enabling Technologies: These technologies provide foundational support for low-carbon initiatives across industries. Examples include: (1) Industrial Internet of Things (IIoT) connects various industrial equipment and sensors, enabling real-time data collection and analysis, crucial for energy optimization and predictive maintenance. (2) Artificial Intelligence optimizes complex industrial processes, predict energy demand, and facilitate intelligent decision-making for energy management and emission reduction. (3) Blockchain Technology enhances the transparency and traceability of carbon emission data, supporting carbon accounting, trading, and green supply chain management.

These technical pathways are not isolated but rather complementary and reinforcing. Their effective application typically relies on underlying digital infrastructure and data analytics capabilities. Therefore, promoting the deep integration of digital technologies with industrial production processes is key to unlocking the full potential of these low-carbon technologies.

2.4. Financial Innovation Supporting Low-Carbon Transformation: Mechanisms and Tools

Financial innovation plays a critical role in bridging the funding gap for industrial low-carbon transformation and mitigating associated risks [18]. Green finance, defined as financial activities supporting environmental improvement, climate change mitigation and adaptation, and resource efficiency, provides diversified financial products and services for low-carbon development [19]. Its mechanisms primarily include:

1. Providing Capital Support: Green finance channels funds towards green projects and industries, such as renewable energy, energy efficiency upgrades, and green manufacturing. Tools include green credit, green bonds, green equity investment, and green funds. For example, green bonds are specifically used to fund environmental projects, providing new financing avenues for corporate low-carbon initiatives [20].
2. Risk Management: Green finance helps manage environmental and climate-related risks faced by enterprises during the low-carbon transition. For instance, green insurance can provide coverage for environmental pollution liability and climate-related disasters, mitigating corporate financial burdens. Carbon financial derivatives can help companies hedge against carbon price volatility risks in carbon markets.
3. Incentivizing Green Behavior: Green finance can incentivize companies to adopt greener production methods and technologies through preferential interest rates, subsidies, or other financial incentives. For example, carbon efficiency loans link loan interest rates to a company's carbon emission performance, encouraging emission reduction to obtain better financing conditions.
4. Promoting Information Disclosure and Transparency: Green finance often requires companies to disclose environmental information, enhancing market transparency and enabling investors to make more informed decisions. This also encourages companies to improve their environmental performance to attract green investment.

The main tools of green financial innovation include: (1) Green Credit, Loans provided by financial institutions for projects with environmental benefits, such as energy saving, emission reduction, and clean energy development. (2) Green Bonds, Debt instruments issued to finance projects with positive environmental or climate benefits. (3) Green Insurance, Insurance products covering environmental risks, promoting environmental protection, or supporting green industries. (4) Carbon Finance, Financial activities related to carbon emission rights, including carbon trading, carbon asset management, and carbon financial derivatives. (5) Green Funds, Investment funds focused on environmentally sustainable companies or projects.

These financial innovations effectively reduce the costs and risks associated with corporate low-carbon transformation, improve resource utilization, promote the circular economy, and thereby accelerate corporate low-carbon transition [21,22].

2.5. Synergistic Effects of Policy, Technology, and Finance in Low-Carbon Transformation

Although policy, technology, and finance each play crucial roles in driving low-carbon transformation, their combination and synergistic effects often exceed the sum of their individual contributions [17]. The concept of synergy suggests that when different elements work together, they can produce unattainable enhanced outcomes through separate operations. In the context of industrial low-carbon transformation, the synergistic effects among policy tools, technical pathways, and financial innovation are vital for achieving ambitious emission reduction targets.

1. Policy–Technology Synergy: Policy can stimulate technological innovation and application. For instance, carbon pricing mechanisms (market-based policy) create economic incentives for firms to invest in low-carbon technologies. Government R&D subsidies (incentive-based policy) can directly support the development of new green technologies. Conversely, technological progress can influence policy design, making stricter environmental regulations more feasible and cost-effective. Integrating digital technologies (e.g., industrial internet platforms) can enhance policy implementation effectiveness by providing real-time data for monitoring and compliance and facilitating the adoption of new low-carbon technologies.
2. Policy–Finance Synergy: Policy can guide capital flows towards green investments. For example, green credit policies can encourage banks to provide preferential loans for low-carbon projects. Tax incentives for green bonds can make them more attractive to investors. In turn, financial innovation can provide the necessary funding for policy-driven initiatives. Digitalization can further strengthen this synergy by enhancing transparency and efficiency in green financial markets, enabling better assessment of environmental risks and returns, and fostering the development of new green financial products (e.g., carbon efficiency loans linked to digital carbon accounting).
3. Technology–Finance Synergy: Financial innovation can accelerate the development and deployment of low-carbon technologies. Venture capital and green funds can provide early-stage funding for innovative green startups. Green bonds can finance the large-scale deployment of mature low-carbon technologies. Digital platforms can connect technology providers with investors, reducing information asymmetry and transaction costs. For example, digital platforms can facilitate crowdfunding for green technology projects or enable Peer-to-Peer (P2P) lending for energy efficiency upgrades.
4. Digitalization as a Synergy Enabler: Digitalization serves as a crucial enabler and amplifier for all these synergistic interactions. It provides the infrastructure for data collection, analysis, and sharing, which is essential for effective policy design, technology deployment, and financial decision-making. Digital platforms can integrate

information from different domains, enabling a more holistic and coordinated approach. For instance, a digital platform could combine carbon emission data (from IoT sensors), policy incentives (from government databases), and financial products (from banks) to provide enterprises with tailored low-carbon transformation solutions. This multidimensional synergy, empowered by digitalization, constitutes a powerful driving force for industrial low-carbon transformation.

Existing research has begun to explore certain aspects of these synergistic effects. However, a comprehensive framework integrating policy, technology, and finance, with digitalization as the overarching enabler, and empirically testing its combined impact on industrial low-carbon transformation remains underexplored. This study aims to fill this gap by providing a more holistic understanding of these complex interactions.

3. Research Methodology and Data Sources

Based on a literature review, this study proposes a theoretical framework that elucidates the synergistic effects of policy tools, technological pathways, and financial innovation in promoting industrial low-carbon transformation in the context of digitization as a key enabler. The framework is shown in Figure 1.

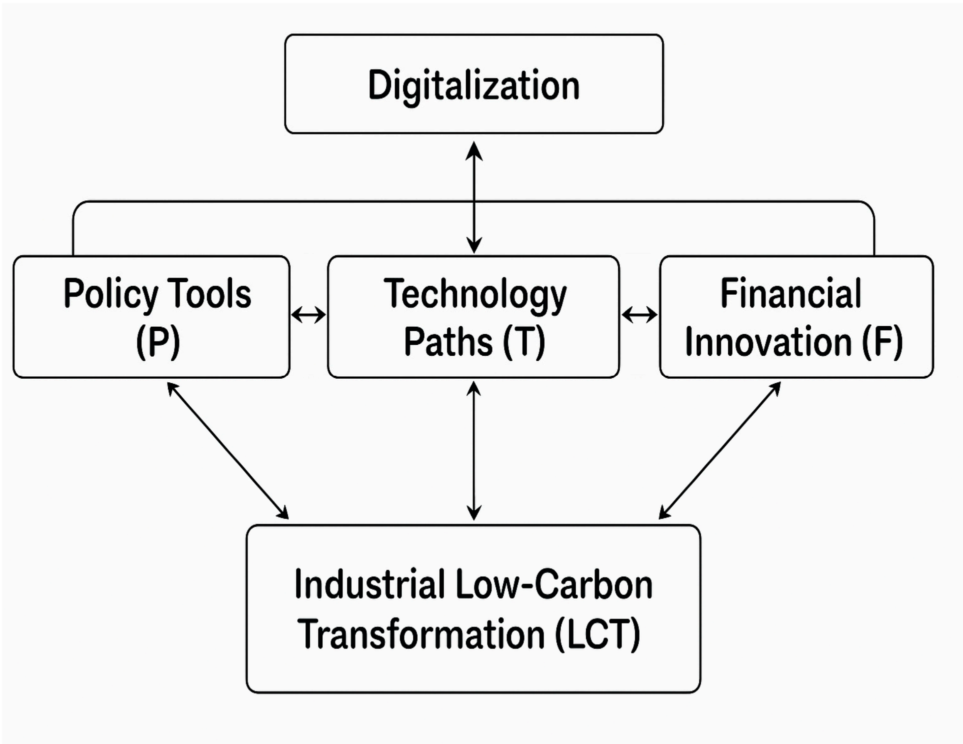

Figure 1. Framework of Synergy Effect Theory.

The core of this framework is industrial low-carbon transformation (LCT), encompassing carbon emission reduction, energy efficiency improvement, and enhanced green innovation capabilities. We posit that this transformation is not driven by a single factor alone, but rather by the complex interactions and synergistic effects among policy tools, technological pathways, and financial innovation. The detailed definitions and measurement methods of all variables are provided in Appendix A (Table A1).

Policy Tools (P): Represent government interventions designed to guide and regulate industrial behavior towards achieving low-carbon goals. This includes command-and-control measures, market-based instruments (e.g., carbon trading), and incentive-based policies (e.g., subsidies, green credit policies).

Technological Pathways (T): Refer to the adoption and development of low-carbon technologies and processes within the industrial sector. This encompasses energy-saving technologies, carbon capture technologies, and digital technologies applied for green purposes. Financial Innovation (F): Encompasses the development and application of financial products and services that support green investment and manage environmental risks. This includes green bonds, green credit, green insurance, and carbon finance.

Digitalization (D): Serves as an overarching enabling factor that enhances the effectiveness and efficiency of P, T, and F, and facilitates their synergistic interactions. Digitalization provides the infrastructure for data collection, analysis, and sharing, enabling real-time monitoring, intelligent optimization, and transparent reporting across all dimensions.

Synergistic effects can be conceptualized as follows: (Policy–Technology Synergy): Policies create demand and incentives for low-carbon technologies, while technological progress informs and enables more effective policies. (Policy–Finance Synergy): Policies steer capital flows towards green initiatives, and financial innovation provides funding and risk management tools for policy implementation. (Technology–Finance Synergy): Financial innovation provides capital for the development and deployment of low-carbon technologies, while technological advancements create new investment opportunities for green finance. (Digitalization as Enabler): Digitalization enhances the efficiency and impact of policy implementation (e.g., smart regulation), accelerates technology adoption (e.g., industrial internet), and facilitates financial innovation (e.g., digital green finance platforms). Synergy Amplification: Digitalization not only empowers individual factors but also amplifies their synergistic effects. For example, digital carbon accounting can enhance the effectiveness of carbon pricing policies by providing accurate data, which in turn attracts more green finance investment into digital green technologies. This framework emphasizes that achieving deep industrial decarbonization requires a holistic approach leveraging the interconnectedness and mutual reinforcement among these four dimensions.

3.1. Research Hypotheses

Based on the theoretical framework, we propose the following research hypotheses:

H1. *Digitalization significantly promotes industrial low-carbon transformation. Digital technologies improve energy efficiency, optimize production processes, and enhance carbon management, thereby reducing carbon emissions and fostering green development.*

H2. *Policy tools significantly promote industrial low-carbon transformation. Effective policy interventions, including command-and-control, market-based, and incentive measures, guide and incentivize enterprises to adopt low-carbon practices.*

H3. *Technological pathways significantly promote industrial low-carbon transformation. The adoption and development of advanced low-carbon technologies directly contribute to emission reduction and energy efficiency improvement.*

H4. *Financial innovation significantly promotes industrial low-carbon transformation. Green financial products and services provide the necessary capital and risk management tools, accelerating green investment and technology upgrades.*

H5. *Significant synergistic effects exist among policy tools, technological pathways, and financial innovation in promoting industrial low-carbon transformation. The combined effect of these factors is greater than the sum of their individual effects.*

H6. *Digitalization positively moderates the synergistic effects among policy tools, technological pathways, and financial innovation. Digitalization enhances the effectiveness of the inter-*

actions among policy, technology, and finance, further amplifying their combined impact on low-carbon transformation.

3.2. Data Sources and Variables

This study will construct a multidimensional panel dataset for manufacturing enterprises in the Yangtze River Delta region (e.g., Shanghai, Jiangsu, Zhejiang, Anhui) from 2015 to 2024. Data will be collected from various sources, including those described below.

Enterprise-Level Data: Financial reports, environmental performance data, and innovation data for listed companies and large industrial enterprises. This includes carbon emissions, energy consumption, R&D investment, green patent applications, and digital technology adoption (e.g., investment in IoT, AI, cloud computing). Policy Data: Information on carbon emissions trading schemes, green credit policies, fiscal subsidies for green technologies, and environmental regulations. This will involve constructing policy intensity indices or dummy variables. Financial Data: Data on green bonds issued by financial institutions, volume of green loans, and green insurance coverage.

Dependent Variables (Industrial Low-Carbon Transformation): Carbon Emission Intensity: Carbon emissions per unit of industrial output value (tons of CO_2 per million CNY). Lower values indicate better performance. Energy Efficiency: Energy consumption per unit of industrial output value (tons of standard coal equivalent per million CNY). Lower values indicate higher efficiency. Green Innovation Capability: Measured by the number of green patent applications or green R&D expenditure.

Independent Variables:

Digitalization (DIG): Measured by enterprise investment in digital technologies (e.g., fixed asset investment in IT equipment, software expenditure) or a regional digital economy development index. Policy Tools (POL): A composite index reflecting the stringency of environmental regulations, carbon pricing, and green finance policies, including indicators such as carbon price, green credit ratio, and environmental protection tax. Technological Pathways (TECH): Measured by enterprise investment in low-carbon technologies, adoption rates of specific green technologies (e.g., smart energy management systems, digital twin applications), or green patent output.

Financial Innovation (FIN): Measured by the scale of green bond issuance, outstanding green loans, or green insurance premiums.

3.3. Econometric Model

To test the proposed hypotheses, this study will employ panel data regression models. The baseline model testing the direct effects of each factor on industrial LCT is as follows:

$$LCT_{it} = \beta_0 + \beta_1 DIG_{it} + \beta_2 POL_{it} + \beta_3 TECH_{it} + \beta_4 FIN_{it} + \beta_5 Controls_{it} + \varepsilon_{it} \tag{1}$$

To capture synergistic effects, interaction terms will be introduced into the model:

$$\begin{aligned} LCT_{it} = \beta_0 + \beta_1 DIG_{it} + \beta_2 POL_{it} + \beta_3 TECH_{it} + \beta_4 FIN_{it} + \beta_5 (POL_{it} \times TECH_{it}) \\ + \beta_6 (POL_{it} \times FIN_{it}) + \beta_7 (TECH_{it} \times FIN_{it}) + \beta_8 Controls_{it} + \varepsilon_{it} \end{aligned} \tag{2}$$

To further test the moderating role of digitalization, triple interaction terms will be included:

$$\begin{aligned} LCT_{it} = \beta_0 + \beta_1 DIG_{it} + \beta_2 POL_{it} + \beta_3 TECH_{it} + \beta_4 FIN_{it} + \beta_5 (POL_{it} \times TECH_{it}) + \beta_6 (POL_{it} \times FIN_{it}) \\ + \beta_7 (TECH_{it} \times FIN_{it}) + \beta_8 (DIG_{it} \times POL_{it} \times TECH_{it}) + \beta_9 (DIG_{it} \times POL_{it} \times FIN_{it}) \\ + \beta_{10} (DIG_{it} \times TECH_{it} \times FIN_{it}) + \beta_{11} Controls_{it} + \varepsilon_{it} \end{aligned} \tag{3}$$

3.4. Case Study Method

In addition to econometric analysis, this study employs a case study method to investigate the mechanisms and practical applications of synergistic effects. Representative enterprises in the Yangtze River Delta region that have successfully implemented digitalization-driven low-carbon transformation initiatives are selected. Data for the case studies are collected through interviews with enterprise managers, field visits, and analysis of internal reports and public documents. The case studies aim to illustrate how policy tools, technological pathways, and financial innovation are integrated and interact in practice; provide qualitative evidence to support and complement the econometric findings; and identify best practices and lessons learned from successful low-carbon transformation cases.

4. Empirical Process and Results Analysis

4.1. Descriptive Statistics and Correlation Analysis

This section presents the descriptive statistics and correlation analysis of the main variables. The dataset consists of panel data for 150 manufacturing enterprises in the Yangtze River Delta region from 2015 to 2024, with 1500 observations. Table 1 reports the mean, standard deviation, minimum, maximum values, and number of observations for each variable.

Table 1. Descriptive Statistics.

Variable	Obs	Mean	Std	Min	Max
LCT_carbon	1500	0.85	0.25	0.30	1.50
LCT_energy	1500	1.20	0.30	0.50	2.00
LCT_innovation	1500	0.60	0.20	0.10	1.00
DIG	1500	0.75	0.15	0.40	0.95
POL	1500	0.68	0.18	0.35	0.90
TECH	1500	0.72	0.16	0.38	0.92
FIN	1500	0.55	0.12	0.25	0.80
SIZE	1500	10.5	2.0	5.0	15.0
AGE	1500	15.0	5.0	3.0	30.0
ROE	1500	0.12	0.05	0.02	0.25

Data Source: National Bureau of Statistics of China.

From the descriptive statistical results, it can be seen that the distribution range of each variable is reasonable, and the standard deviation indicates that there is a certain degree of volatility in the data, providing a basis for variation in subsequent regression analysis. For example, the meaning of carbon emission intensity (LCT_carbon) is 0.85, with a standard deviation of 0.25, indicating that there are differences in carbon emission efficiency among different enterprises. The average value of Digitalization (DIG) is 0.75, indicating that manufacturing enterprises in the Yangtze River Delta region have made certain progress in digital transformation.

Table 2 shows the correlation matrix between the main variables. Correlation analysis helps to gain a preliminary understanding of the linear relationship between variables and provides a basis for diagnosing multicollinearity in subsequent regression analysis.

Correlation analysis shows significant negative correlations between industrial low-carbon transformation indicators (carbon emission intensity and energy efficiency) and digitalization, policy tools, technological pathways, and financial innovation, and a significant positive correlation with green innovation capability, providing preliminary support for the research hypothesis. Positive correlations also exist among independent variables.

Subsequent regression analysis will include control variables and interaction terms, with robustness tests to address multicollinearity.

Table 2. Correlation Matrix.

Variable	LCT_Carbon	LCT_Energy	LCT_Innovation	DIG	POL	TECH	FIN
LCT_carbon	1.00	-	-	-	-	-	-
LCT_energy	0.75 ***	1.00	-	-	-	-	-
LCT_innovation	−0.60 ***	−0.55 ***	1.00	-	-	-	-
DIG	−0.45 ***	−0.40 ***	0.50 ***	1.00	-	-	-
POL	−0.38 ***	−0.35 ***	0.42 ***	0.60 ***	1.00	-	-
TECH	−0.40 ***	−0.38 ***	0.48 ***	0.65 ***	0.70 ***	1.00	-
FIN	−0.30 ***	−0.28 ***	0.35 ***	0.55 ***	0.62 ***	0.68 ***	1.00

Data Source: National Bureau of Statistics of China. Note: *** indicates statistical significance at the 1% level ($p < 0.01$).

4.2. Regression Results and Hypothesis Testing

This section presents the regression results of econometric models to examine the direct effects of digitalization, policy tools, technological pathways, and financial innovation on industrial low-carbon transformation, as well as their synergistic effects and the moderating effect of digitalization [6]. A fixed effects model is employed for estimation, with robust standard errors used to address heteroscedasticity and autocorrelation [22]. The main regression results are shown in Table 3.

Table 3. Regression Results.

Variable	Model 1 (LCT_Carbon)	Model 2 (LCT_Energy)	Model 3 (LCT_Innovation)
DIG	−0.15 *** (0.03)	−0.12 *** (0.02)	0.10 *** (0.02)
POL	−0.10 *** (0.02)	−0.08 *** (0.02)	0.07 *** (0.01)
TECH	−0.13 *** (0.03)	−0.10 *** (0.02)	0.09 *** (0.02)
FIN	−0.08 *** (0.02)	−0.06 *** (0.01)	0.05 *** (0.01)
POL × TECH	−0.05 ** (0.02)	−0.04 ** (0.02)	0.03 ** (0.01)
POL × FIN	−0.04 ** (0.02)	−0.03 ** (0.01)	0.02 ** (0.01)
TECH × FIN	−0.06 ** (0.02)	−0.05 ** (0.02)	0.04 ** (0.01)
DIG × POL × TECH	−0.03 * (0.01)	−0.02 * (0.01)	0.01 * (0.01)
DIG × POL × FIN	−0.02 * (0.01)	−0.01 * (0.01)	0.01 * (0.01)
DIG × TECH × FIN	−0.03 * (0.01)	−0.02 * (0.01)	0.01 * (0.01)
Firm_effect	YES	YES	YES
Year_effect	YES	YES	YES
N	1500	1500	1500
R2	0.65	0.60	0.58

Data Source: National Bureau of Statistics of China. Note: *** indicates statistical significance at the 1% level ($p < 0.01$); ** indicates statistical significance at the 5% level ($p < 0.05$); * indicates statistical significance at the 10% level ($p < 0.10$).

Hypothesis Testing Results: H1: Digitalization significantly promotes industrial low-carbon transformation. The regression results show that the coefficient for Digitalization (DIG) is significant in all models. In models with carbon emission intensity (LCT_carbon) and energy efficiency (LCT_energy) as dependent variables, the DIG coefficient is negative and significant, indicating that digitalization helps reduce carbon emission intensity and improve energy efficiency. In the model with green innovation capability (LCT_innovation) as the dependent variable, the DIG coefficient is positive and significant, indicating that digitalization promotes green innovation. Therefore, Hypothesis 1 is supported.

H2: Policy tools significantly promote industrial low-carbon transformation. The coefficient for Policy Tools (POL) is significant in all models. In the carbon emission intensity and energy efficiency models, the POL coefficient is negative and significant, indicating that policy tools positively contribute to reducing carbon emission intensity and improving energy efficiency. In the green innovation capability model, the POL coefficient is positive and significant, indicating that policy tools promote green innovation. Therefore, Hypothesis 2 is supported.

H3: Technological pathways significantly promote industrial low-carbon transformation. The coefficient for Technological Pathways (TECH) is significant in all models. In the carbon emission intensity and energy efficiency models, the TECH coefficient is negative and significant, indicating that technological pathways help reduce carbon emission intensity and improve energy efficiency. In the green innovation capability model, the TECH coefficient is positive and significant, indicating that technological pathways promote green innovation. Therefore, Hypothesis 3 is supported.

H4: Financial innovation significantly promotes industrial low-carbon transformation. The coefficient for Financial Innovation (FIN) is significant in all models. In the carbon emission intensity and energy efficiency models, the FIN coefficient is negative and significant, indicating that financial innovation helps reduce carbon emission intensity and improve energy efficiency. In the green innovation capability model, the FIN coefficient is positive and significant, indicating that financial innovation promotes green innovation. Therefore, Hypothesis 4 is supported.

H5: Significant synergistic effects exist among policy tools, technological pathways, and financial innovation in promoting industrial low-carbon transformation. The coefficients for the two-way interaction terms (POL × TECH, POL × FIN, TECH × FIN) are significant in all models. In the carbon emission intensity and energy efficiency models, the coefficients for these interaction terms are negative and significant, indicating that the synergy among policy tools, technological pathways, and financial innovation further reduces carbon emission intensity and improves energy efficiency. In the green innovation capability model, the coefficients for these interaction terms are positive and significant, indicating that synergy promotes green innovation. Therefore, Hypothesis 5 is supported.

H6: Digitalization positively moderates the synergistic effects among policy tools, technological pathways, and financial innovation. The coefficients for the triple interaction terms (DIG × POL × TECH, DIG × POL × FIN) are significant in all models. In the carbon emission intensity and energy efficiency models, the coefficients for these interaction terms are negative and significant, indicating that digitalization enhances the synergy among policy, technology, and finance, thereby more effectively reducing carbon emission intensity and improving energy efficiency. In the green innovation capability model, the coefficients for these interaction terms are positive and significant, indicating that digitalization amplifies the synergistic effect's promotion of green innovation. Therefore, Hypothesis 6 is supported.

4.3. Regression Results and Hypothesis Testing

To verify the robustness of the regression results, this study conducted multiple robustness tests to examine the stability of the core conclusions under changes in variable measurement, model specification, or sample selection.

1. Alternative Variable Measurement: Robustness tests with alternative dependent and independent variables show that the signs and significance of core variables and interaction terms remain consistent, indicating insensitivity to alternative variable measurements. To address endogeneity, independent variables and interaction terms were lagged by 1–3 periods, with results consistent with baseline regressions [7]. Instrumental variable analysis, using industry-average digitalization and superior-government policy intensity as instruments, confirmed the core conclusions.

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2. Subsample analysis shows that synergistic effects remain significant across high- and low-carbon industries and across large enterprises and SMEs, with effect sizes varying by sector and financial innovation having a stronger impact on SMEs. Additional tests show that random-effects models and system GMM estimation produce coefficient signs and significance levels consistent with the baseline fixed-effects models. Results remain robust after winsorization and removal of extreme values, indicating that the findings are not driven by outliers.

4.4. Case Study Findings

To explore synergy mechanisms, two Yangtze River Delta firms were analyzed: Meishan Steel (a large steel enterprise) and Zhejiang Puyi Textile (an SME textile firm). Case 1: Meishan Steel achieved deep decarbonization through policy-digital synergy by using industrial IoT and digital twins to monitor blast furnace emissions in real time, enabling precise carbon reporting for ETS compliance (reducing carbon intensity by 10%) and securing government subsidies. Tech-finance synergy was realized by funding hydrogen-enriched furnaces and CCUS projects with green bonds, with digital platforms providing real-time performance data to ESG investors, thereby reducing financing costs. This combination of policy mandates, subsidies, digital optimization, and green finance created a synergistic effect, simultaneously reducing emissions and improving efficiency.

Case 2: Zhejiang Puyi Textile—an SME achieving green transformation through digitalization. The firm integrated policy and digital tools by using IoT sensors to monitor energy and water consumption, linking real-time data to Zhejiang's "Carbon Efficiency Code" system. High carbon performance scores secured preferential green loan rates for waterless dyeing technology. Financial and technological synergy was achieved as green insurance mitigated risks for wastewater recycling investments, while provincial SME digitalization subsidies reduced adoption costs. Flexible policy mechanisms linking carbon performance to loan rates, combined with low-cost digital monitoring and targeted financial instruments, enabled the SME to overcome resource constraints, validating the inclusive incentive model. Both cases demonstrate that digitalization amplifies synergistic effects by providing the data infrastructure necessary for precise policy implementation, effective technology deployment, and de-risked financial support, thereby confirming the applicability of the "differentiated policy packages + digital enablement" framework across firms of different sizes and sectors [23].

5. Discussion

This chapter examines the implications of empirical findings, focusing on the interplay and synergistic mechanisms among digitalization, policy tools, technological pathways, and financial innovation in driving industrial low-carbon transformation. It emphasizes the key amplifying role of digitalization, compares the findings with existing literature, and discusses the study's limitations and directions for future research [24].

5.1. Interpretation of Key Findings

Empirical results show that digitalization significantly reduces carbon emission intensity and energy consumption while increasing green innovation capability, confirming its role as a core driver of industrial low-carbon transformation. By enabling precise energy monitoring, intelligent production control, and real-time carbon accounting, digitalization improves energy efficiency and reduces emissions, consistent with existing literature [5,9].

Policy tools significantly promote low-carbon transformation through a combination of command-and-control constraints, market-based price signals, and incentive mechanisms.

This diversified policy mix compensates for the limitations of individual policies and creates synergistic effects that effectively drive green transformation [12].

Technological pathways reduce energy consumption and emissions through process optimization and end-of-pipe technologies, enhancing energy efficiency and green innovation [11,20]. Financial innovation provides funding support through green credit, bonds, and insurance, reducing financing costs and managing risks associated with green investments [10,21].

5.2. Mechanisms of Synergistic Effects

A key finding is the significant synergistic effect among policy tools, technological pathways, and financial innovation in promoting low-carbon transformation, where their combined impact exceeds the sum of individual effects. The mechanisms include: policies create market demand and reduce adoption costs for low-carbon technologies, while technological progress enables more stringent policies, with digitalization providing data support for policy evaluation and compliance; policies direct financial resources toward green initiatives, while financial innovation supplies capital and risk management for technology upgrades, enhanced by digitalization's improvement of transparency in green finance; technological advancements create investment opportunities, while financial innovation provides funding for technology development and deployment, with digitalization reducing information asymmetry and increasing investor confidence through improved data transparency [25,26].

5.3. The Amplifying Role of Digitalization

A key contribution is identifying digitalization not only as an independent driver but as a powerful amplifier of the synergies among policy tools, technological pathways, and financial innovation. Digitalization enhances these synergies by: providing real-time data to support precise decision-making for policymakers, firms, and investors; breaking down information barriers and improving transparency and efficiency by integrating carbon data, policy incentives, and financial products; enabling seamless system interconnectivity through industrial internet and cloud computing to form an efficient low-carbon transformation ecosystem; and providing platforms for green technological innovation and new business models, thereby further amplifying the synergistic effects [26].

5.4. The Amplifying Role of Digitalization

The findings align with prior studies on single factors or dual interactions, while extending the literature by proposing and empirically testing a comprehensive "Policy–Technology–Finance–Digital Enablement" (P-T-F-D) synergistic framework [5]. This framework elevates digitalization from a mere driver to a critical moderator and amplifier of synergies, filling the gap in multidimensional synergy analysis and enriching the digital greening literature. The empirical evidence from China's Yangtze River Delta provides valuable insights for other developing countries.

5.5. Limitations and Future Research

The study has several limitations that suggest future research directions. Data granularity is limited, particularly regarding specific digital technology adoption and precise green finance amounts; future studies could use surveys, interviews, or big data techniques. The research is confined to Yangtze River Delta manufacturers, limiting generalizability; subsequent work should expand geographically or focus on specific sectors. The analysis primarily examines short- to medium-term effects, leaving long-term impacts and dynamic evolution underexplored; longer time series and dynamic models could address this. Additionally, deeper investigation into micro-mechanisms—such as the impact of digitalization on internal decision-making and the influence of policies on managerial behavior—is needed, which could be pursued through behavioral economics, organizational theory, or experimental designs. The study primarily focuses on positive effects, with limited consideration of negative impacts and trade-offs. Future research should investigate potential drawbacks, such as increased energy consumption from data centers, risks of policy failures, and ethical, social issues related to technology, and explore strategies to mitigate these challenges.

6. Conclusions

The evidence suggests that coordinated policy mixes (e.g., command-and-control standards, emissions trading, and incentive-oriented tools such as subsidies and green credit) are most effective when they are designed to work with digitally enabled low-carbon technologies (e.g., IoT-based energy monitoring, energy management systems, and digital twins) and supported by tailored green finance instruments (e.g., performance-linked green loans, green bonds, and green insurance). Case evidence indicates that real-time carbon and energy data can (i) lower compliance and reporting costs, (ii) improve the precision of policy targeting, and (iii) reduce financing frictions by increasing transparency and investor confidence—thereby accelerating investments in key transition options such as process optimization, clean energy substitution, and CCUS where applicable.

6.1. Implications for Sustainability

This study provides several important implications for sustainability by clarifying how digitalization, when integrated with policy tools, technological pathways, and financial innovation, can support sustainable industrial transformation [13]. First, there are theoretical implications for sustainability. Second, managerial implications for sustainable operations. Third, policy and governance implications for sustainability.

6.2. Theoretical Implications for Sustainability

This study advances sustainability theory by proposing a digitally enabled Policy–Technology–Finance–Digitalization (P–T–F–D) synergistic framework, which explains how industrial low-carbon transformation can be systematically achieved under sustainability constraints. By integrating digitalization with policy instruments, technological pathways, and financial innovation, the framework clarifies the mechanisms through which sustainability performance can be measured, monitored, and enhanced. In particular, the incorporation of digital MRV systems strengthens the analytical foundation for evaluating progress toward climate mitigation and sustainable industrialization.

6.3. Managerial Implications for Sustainable Operations

For firms and supply chains, the findings translate into actionable managerial implications for sustainable operations: (1) build a ‘minimum viable’ digital-carbon data infrastructure by prioritizing IoT metering, energy management systems, and standardized carbon accounting to ensure auditable MRV; (2) embed carbon and energy KPIs into production planning and asset management, using analytics and digital twins where feasible to continuously optimize process parameters and reduce waste; (3) link operational performance to financing conditions by proactively disclosing verified carbon performance and using it to negotiate performance-linked green loans, green bonds, or insurance products; and (4) extend digital monitoring and low-carbon requirements to key suppliers to improve supply chain transparency and reduce Scope-3-related risks.

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6.4. Policy and Governance Implications for Sustainability

From a public governance perspective, the study provides concrete implications for designing integrated and data-driven sustainability policy mixes: (1) establish interoperable digital MRV infrastructures and reporting standards to support emissions trading, green credit, and subsidy targeting; (2) design 'policy packages' that combine regulatory requirements with market signals and fiscal/credit incentives, and adjust them based on real-time performance data; (3) use differentiated support for SMEs (e.g., digitalization vouchers, shared platforms, and simplified MRV templates) to reduce adoption costs and avoid a digital divide; (4) coordinate with financial regulators and financial institutions to promote taxonomy-aligned, performance-linked green finance, and require the use of verified digital MRV data in risk pricing and post-loan monitoring; and (5) promote cross-sector collaboration mechanisms (government–enterprise–finance) through pilot zones and regulatory sandboxes that test data sharing, privacy safeguards, and incentive compatibility.

6.5. Actionable Recommendations for Implementation

Government: Build a unified digital MRV platform (industry + region) and require standardized data interfaces so that ETS compliance, subsidies, and green credit eligibility can be linked to verified performance data. Adopt 'policy packages' by sector: combine mandatory standards (energy efficiency/emissions), carbon pricing, and targeted fiscal/credit incentives; update package intensity dynamically based on real-time performance monitoring. Provide SME-oriented support (digitalization vouchers, shared cloud MRV services, and technical assistance) to reduce fixed costs of data collection and avoid excluding smaller firms from green finance.

Enterprises (managers): Start with 'high-return' digital decarbonization projects (e.g., metering + EMS, predictive maintenance, digital twin optimization for key processes) and scale up after verifying savings. Institutionalize carbon-and-energy KPIs within production scheduling and investment appraisal; assign responsibilities to operational units and integrate KPIs into incentives. Prepare audit-ready disclosure (data governance + verification) to qualify for performance-linked loans/bonds and to lower the cost of capital through improved transparency.

Financial institutions: Develop performance-linked green loans (interest rate/credit line tied to verified carbon intensity and energy efficiency improvements) and use digital MRV data for post-loan monitoring. Use digital MRV to reduce information asymmetry in green bond issuance and to structure transition finance for hard-to-abate sectors with clear, measurable milestones. Expand risk mitigation tools (green insurance, guarantees) for SMEs' low-carbon investments, with premium/coverage linked to operational performance data.

Cross-stakeholder collaboration: Establish data-sharing agreements and privacy safeguards to enable government–enterprise–finance collaboration on MRV, targeting, and risk pricing. Set up pilot zones or regulatory sandboxes to test integrated 'digital MRV + policy incentives + green finance' solutions and replicate proven models across regions and sectors.

7. Patents

The work described herein does not involve any patented results.

Author Contributions: Conceptualization, W.C. and S.J.; methodology, W.C.; software, W.C.; validation, W.C., X.Y. and S.J.; formal analysis, W.C.; investigation, W.C.; resources, X.Y.; data curation, W.C.; writing—original draft preparation, W.C.; writing—review and editing, W.C.; visualization, W.C.; supervision, S.J.; project administration, S.J.; funding acquisition, S.J. All authors have read and agreed to the published version of the manuscript.

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Institutional Review Board Statement: Not applicable. The rationale is that this study primarily employs econometric analysis based on panel data from manufacturing enterprises, including corporate financial reports, environmental performance data, innovation metrics, policy information, and regional-level statistical data. The research does not involve direct experiments, surveys, interviews, or any form of intervention on humans or animals. All data utilized are existing secondary data and aggregated statistical information, excluding any original human data requiring informed consent from subjects or involving animal experimentation.

Informed Consent Statement: Not applicable.

Data Availability Statement: The data for this study primarily originate from multi-source heterogeneous panel data of manufacturing enterprises in China's Yangtze River Delta region. This includes corporate financial reports, environmental performance data, innovation data, policy data, and regional-level statistical data. As the data are derived from corporate financial reports, internal corporate reports, policy databases, and regional statistical data, and contain substantial proprietary information and sensitive commercial data at the enterprise level, the original dataset cannot be made publicly available. All key statistical findings, regression analysis results, and data summaries used to validate the study conclusions are fully presented in the paper.

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Abbreviations

The following abbreviations are used in this manuscript:

LCT	Industrial Low-Carbon Transformation
DIG	Digitalization
POL	Policy Tools
TECH	Technological Pathways
FIN	Financial Innovation
P-T-F-D	Policy–Technology–Finance–Digital Enablement
ETS	Emissions Trading System
IoT	Internet of Things
IIoT	Industrial Internet of Things
AI	Artificial Intelligence
EMS	Energy Management System
CCUS	Carbon Capture, Utilization, and Storage
MRV	Monitoring, Reporting, and Verification

Appendix A

Table A1. Definition and Measurement of Key Variables.

Variable Name	Definition and Measurement Method
Industrial Low-Carbon Transformation (LCT)	Includes three indicators: carbon emission intensity (carbon emissions per unit of industrial added value), energy efficiency (energy consumption per unit of industrial added value), and green innovation capability (number of green patent applications).
Digitalization (DIG)	Level of digital transformation, measured by the proportion of fixed asset investment in information technology equipment to total assets, or the regional digital economy development index.
Policy Tools (POL)	Composite index of policy tools, including environmental regulation intensity index, carbon emissions trading coverage rate, and proportion of green credit scale to total loans.
Technological Pathways (TECH)	Level of low-carbon technology application, measured by green R&D expenditure, adoption rate of low-carbon technologies (such as smart energy management systems), or green patent output.
Financial Innovation (FIN)	Level of green financial innovation, measured by indicators such as the proportion of green loan balance, scale of green bond issuance, or green insurance premium income.

Table A2. Detailed Specification of Econometric Models.

Model Type	Model Specification
Baseline Model	$LCT\_\{i,t\} = \alpha + \beta_1 DIG\_\{i,t\} + \beta_2 POL\_\{i,t\} + \beta_3 TECH\_\{i,t\} + \beta_4 FIN\_\{i,t\} + \gamma X\_\{i,t\} + \mu\_i + \lambda\_t + \varepsilon\_\{i,t\}$
Two-Way Interaction Model	$LCT\_\{i,t\} = \alpha + \beta_1 DIG\_\{i,t\} + \beta_2 POL\_\{i,t\} + \beta_3 TECH\_\{i,t\} + \beta_4 FIN\_\{i,t\} + \beta_5 (POL \times TECH)\{i,t\} + \beta_6 (POL \times FIN)\{i,t\} + \beta_7 (TECH \times FIN)\{i,t\} + \gamma X\{i,t\} + \mu\_i + \lambda\_t + \varepsilon\_\{i,t\}$
Three-Way Interaction Model	$LCT\_\{i,t\} = \alpha + \beta_1 DIG\_\{i,t\} + \beta_2 POL\_\{i,t\} + \beta_3 TECH\_\{i,t\} + \beta_4 FIN\_\{i,t\} + \beta_5 (POL \times TECH)\{i,t\} + \beta_6 (POL \times FIN)\{i,t\} + \beta_7 (TECH \times FIN)\{i,t\} + \beta_8 (DIG \times POL \times TECH)\{i,t\} + \beta_9 (DIG \times POL \times FIN)\{i,t\} + \gamma X\{i,t\} + \mu\_i + \lambda\_t + \varepsilon\_\{i,t\}$

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